A01 Self-Reflection Journal:

GitHub and Colab Lab

Michael Rivas

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Patricia McManus

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What I Did:

This lab involved working with two main tools: GitHub and Jupyter Notebooks. The overall experience centered on building familiarity with how coding projects are created, organized, and shared using real development tools.

On GitHub, I gained exposure to how repositories function as structured spaces for managing projects. Creating a repository and working within it helped me understand that coding work is not just about writing scripts, but also about organizing and tracking changes over time. The idea of commits stood out as a way of preserving progress and maintaining a clear history of work.

In parallel, I worked with a Jupyter Notebook environment through Google Colab. This introduced a more interactive way of coding, where I could combine explanations and Python code in one place. The ability to run small sections of code and immediately see results made the learning process more engaging and easier to follow.

Finally, I used GitHub Desktop to connect my local work with my online repository. This step highlighted how files move between local and cloud environments, and how tools like GitHub help keep everything synchronized.

What I Learned:

This lab helped me understand that programming is not only about writing code, but also about managing and organizing work effectively. GitHub introduced me to the concept of version control, which ensures that every change is recorded. This made me realize how important structure and organization are in real-world software development and collaborative projects.

I also learned how Jupyter Notebooks support a more flexible and interactive style of programming. Unlike traditional coding environments, notebooks allow for a mix of explanation and execution. This made it easier to understand Python basics because I could immediately see the output of simple commands like

```
print("Hello, World!").
```

Another important takeaway was understanding how different tools work together in a workflow. GitHub manages storage and version history, Jupyter provides the coding environment, and GitHub Desktop acts as the bridge between local and online work. Seeing how these tools connect helped me better understand how developers manage real projects.

Challenges Encountered

One challenge was understanding how local files in GitHub Desktop connect to the online repository. At first, it was confusing why changes were not appearing on GitHub even though I had added the file. This was resolved once I better understood how the local repository folder must match the correct project directory.

Another challenge was managing multiple platforms at once. Switching between GitHub, Jupyter (Google Colab), and GitHub Desktop made the workflow feel slightly fragmented at the beginning. It took some adjustment to understand how each tool fits into the overall process.

Questions or Comments

One question I have is how professionals manage much larger projects with many files and collaborators without losing track of changes. I understand the basic idea of commits and repositories, but I am curious about how version control scales in real-world software teams.

Another comment is that Jupyter Notebooks felt very beginner-friendly and helpful for learning Python. The combination of text and code made it easier to follow along compared to traditional coding environments. It would be interesting to explore more advanced uses of notebooks in data analysis or machine learning in the future.

The following link is where my profile's GitHub Repo is located:
https://github.com/mrivassnj-svg/jupyter-exploration